ShockValue

IoT-powered household energy intelligence for clearer bills, safer demand decisions, and a more responsive local grid.

Meet the team,

ShockValue!



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The Problem

01 A total usage reading leaves the household asking what caused the demand.

02 Circuit activity, cost estimates, and usage changes need a shared context.

03 Households and field teams need a clear next step when demand rises.

04 Any grid insight must also show whether the underlying readings can be trusted.

The Solution — Introducing ShockValue

A decision layer that connects circuit readings, demand, estimated cost, and understandable action.

01 Observe total household demand and mapped breaker channels.

02 Compare demand with a configured local operating envelope.

03 Explain likely usage and cost impacts in plain language.

04 Keep readings and human explanations together for review.

What is ShockValue?

Observe

Household demand and breaker-channel activity.

Understand

Usage trends, estimated bills, and unmapped load.

Anticipate

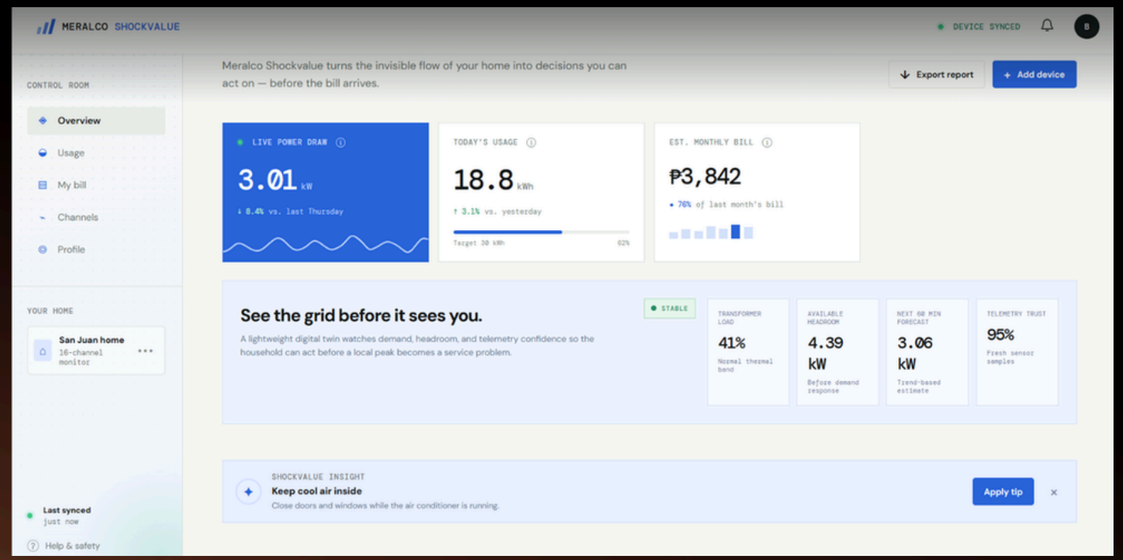
A 60-minute forecast and possible local grid stress.

Act

Energy-saving insight and exportable reports.

The Live Dashboard

A responsive overview answers:
“What is happening right now?”



01 Total power draw in kW and estimated daily energy in kWh.

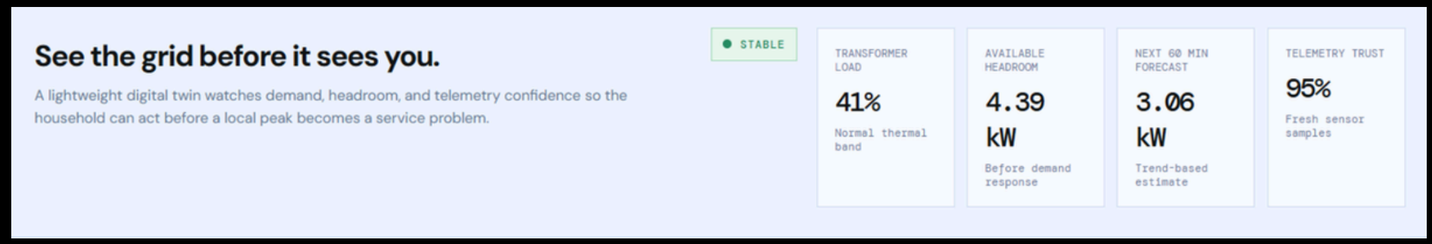
02 Estimated monthly bill with live breaker readings.

03 Grid Intelligence panel and plain-language energy insight.

04 Information buttons, notifications, and text report export.

Grid Intelligence

A lightweight digital-twin-style view makes demand and local constraints easier to understand.



Transformer Load: estimated operating-envelope utilization.

Next 60-Minute Forecast: trend-based projected demand.

Available Headroom: capacity before demand response is suggested.

Telemetry Trust: freshness and consistency; status: STABLE or WATCH.

Breaker Channels

Label monitored circuits so the household can connect readings with a room or known load.

01 Add and select mapped breaker channels through CH1 to CH16.

02 Name an appliance or load and assign a room or circuit.

03 Edit labels or remove a mapping at any time.

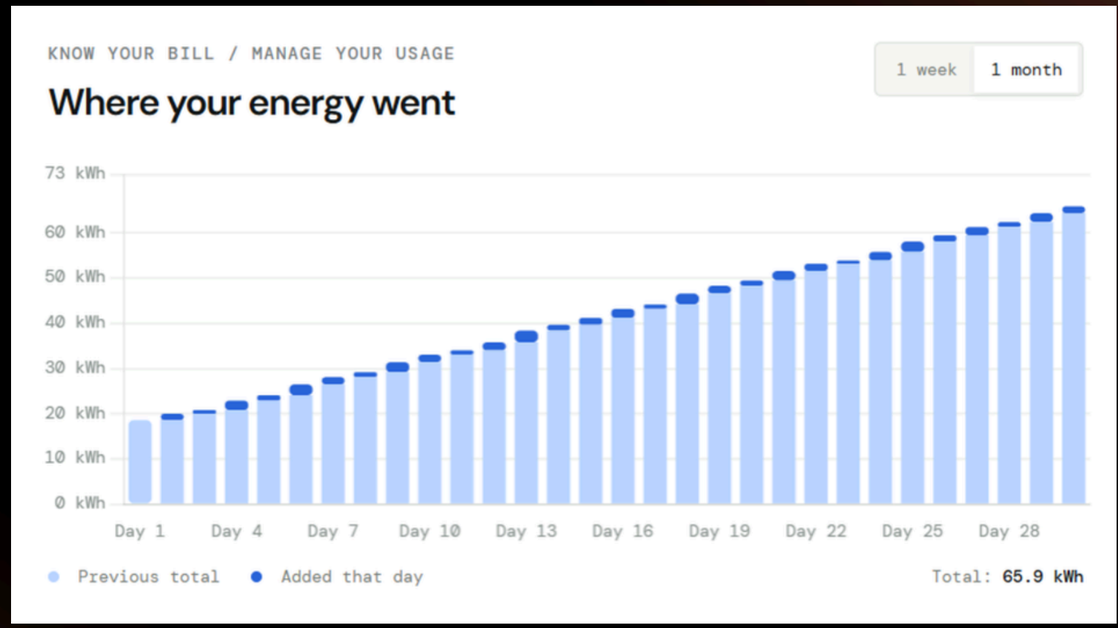
Usage

Review how consumption changes over time and connect patterns with household activity.

01 Choose 7, 30, or 90 day views.

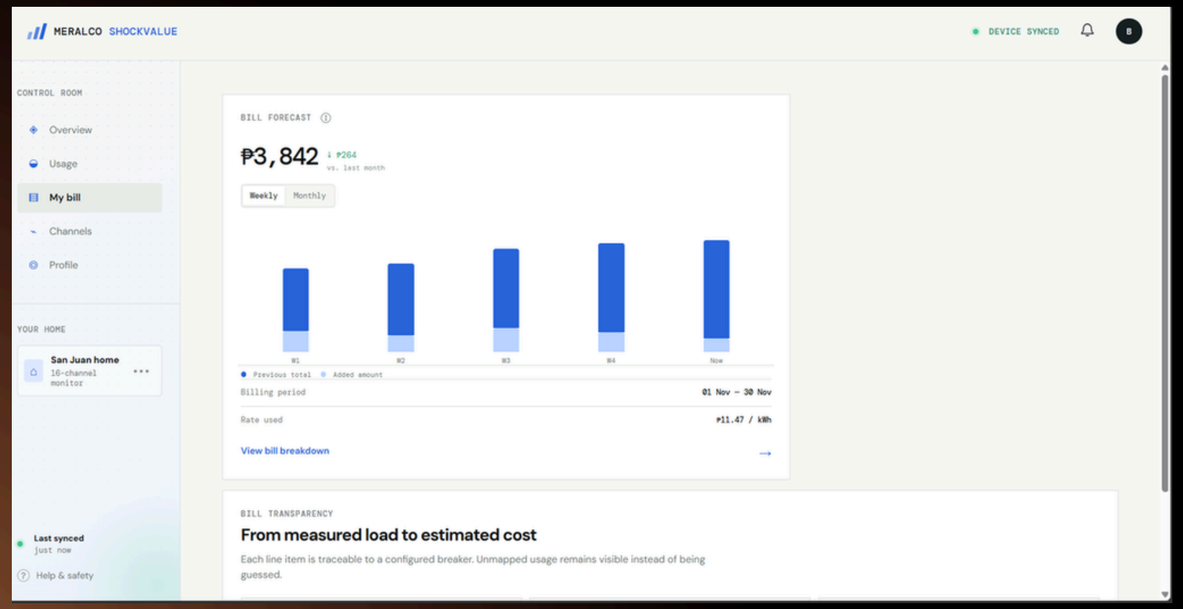
02 View actual-usage history with automatic chart-range switching.

03 Compare with a daily-average reference and peak indicator.



My Bill

Understand estimated expenses and the assumptions behind them.



01 See the estimated billing-period amount and rate assumption.

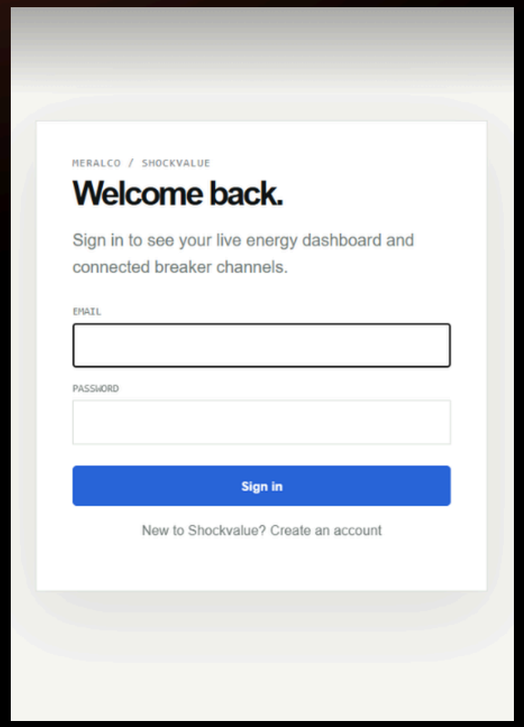
02 Review the bill trend visualization.

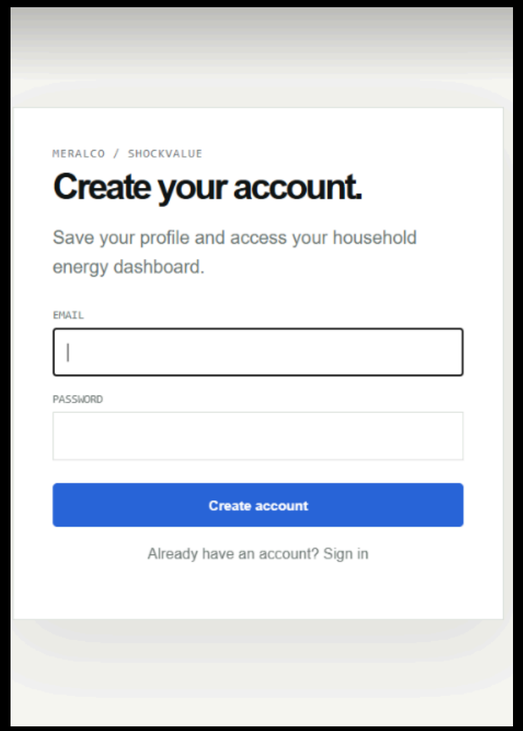
03 Open the circuit-level breakdown and measured-circuit cost view.

04 See the unmapped remainder alongside a clear estimate disclaimer.

Accounts & Profile

A Laravel login flow connects the dashboard with a signed-in user.





01 Public visitors start at login, with a separate signup page.

02 Login or signup opens the dashboard; logout ends the session.

03 Edit username; UserID stays read-only. Verify current password to change email.

04 Password changes require the current password and confirmation; logout ends the session.

How ShockValue Works — Decision Flow

01 Observe

Read household demand and configured breaker channels.

02 Compare

Check expected demand, headroom, and telemetry quality.

03 Interpret

Show possible peaks and their usage or cost impact.

04 Recommend

Suggest a clear next step; retain context for review.

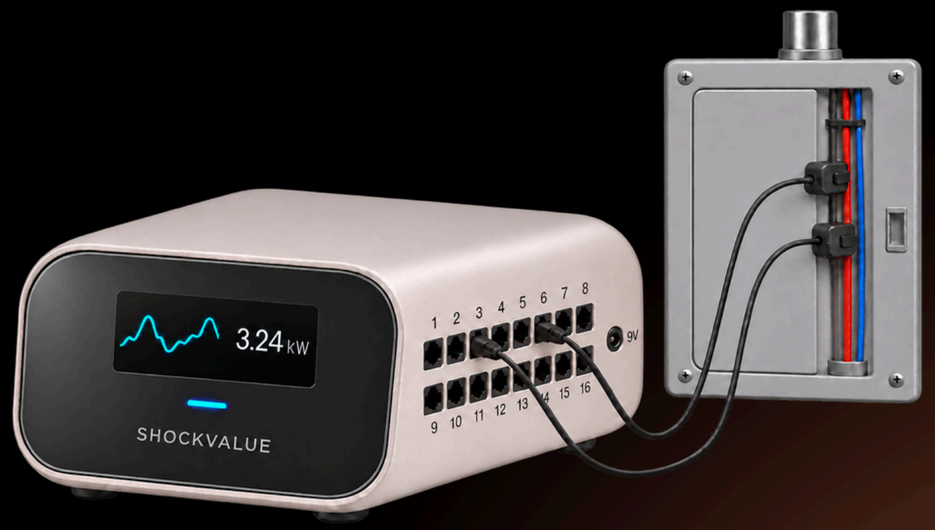
How the Device Would Work

01 Sense: appropriately installed sensors monitor electrical activity on selected circuits.

02 Process: an approved microcontroller collects the quantities needed for power and energy.

03 Transmit: a planned MQTT or HTTPS connection sends authenticated readings.

04 Display: the app presents demand, circuit activity, history, and estimated cost.



Application & Data

Laravel handles accounts. Browser storage supports the dashboard demonstration.





01 SQLite users table: account fields, password hashes, and timestamps.

02 Laravel sessions table: server-side login sessions.

03 Browser localStorage: simulated readings, mappings, history, and notes.

Device Architecture & Security

Sense at the edge

An ESP32/ESP8266 can aggregate CT-clamp readings. Use isolated or utility voltage data for real-power estimates.

Secure transport

Sends telemetry over MQTT/TLS or HTTPS and deploy signed OTA updates.

Validate ingestion

Check identity, timestamp, sequence, units, and signature; buffer messages and apply rate limits before Laravel.

Prototype status

Today, readings are simulated in browser storage. No live gateway or IoT ingestion service is connected.

Safety & Responsible Use

01 Use ShockValue for monitoring and demonstration; it is not a certified protection device.

02 Only a licensed electrician or qualified instructor should install sensors in a breaker panel.

03 Never connect the prototype directly to the main channels or modify a mains cable for a demo.

CONCLUSION

ShockValue connects household demand, breaker-channel activity, estimated cost, and telemetry quality in one understandable view.

Its goal is to help households and field teams understand a change in electricity use and choose a sensible next step.



System Demonstration

Thank You!

ShockValue